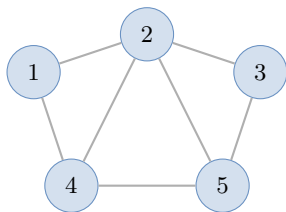


Grafo $G = (V, E)$



Matriz de Adyacencia A

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

Matriz de Grado D

$$\begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}$$

Laplaciano $L = D - A$

$$\begin{pmatrix} \mathbf{2} & -1 & 0 & -1 & 0 \\ -1 & \mathbf{4} & -1 & -1 & -1 \\ 0 & -1 & \mathbf{2} & 0 & -1 \\ -1 & -1 & 0 & \mathbf{3} & -1 \\ 0 & -1 & -1 & -1 & \mathbf{3} \end{pmatrix}$$